
DRAGOON MOUNTAINS GROWING GUIDES

pH Up & pH Down Replacements

Locally-Foraged and Homemade Alternatives for Dragoon Mountain Hydroponic Growers

Dragoon Mountains Growing Guides

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This is the acid-and-alkali companion to the Dragoon Mountains library. Pair it with:

- **A — Pepper Growing Guide** — pH targets for peppers (5.8-6.2)
- **B — Bato Bucket Hydroponic Setup** — the build that most of these recipes are dosed into
- **D — A Hydroponic Primer** — water treatment and the bicarbonate scrub in context
- **E — Modern Masterblend Guide** — the commercial pH Down / pH Up this guide replaces
- **F — Hydroponic Nutrients Guide** — the locally-sourced nutrient programme these acids dose into

Why pH Control Matters

Whether you're using SE Arizona well water or collected monsoon rainwater, pH management is the most important daily task in hydroponics. Nutrients become chemically unavailable outside the ideal range — plants can starve in a perfectly mixed reservoir if the pH is wrong.



Figure 1 – The Dragoon Mountains rising above the Sulphur Springs Valley, Cochise County, Arizona.

pH range	What happens	Common in the Dragoons?
Below 5.5	Iron, manganese, copper toxicity. Root damage.	Rare — only if overdosing acid
5.8 - 6.2 ✓ Target	All nutrients fully available. Optimal for most crops.	Requires treatment of local water
6.5 - 7.0	Iron and manganese start locking out. Yellowing begins.	Common with well water — correct it
7.5 - 8.5	Most micronutrients locked out. Serious deficiencies.	Very common with untreated SE AZ well water

The Bicarbonate Problem in SE Arizona

SE Arizona well water is rich in bicarbonates — dissolved minerals that act as a pH buffer, constantly pushing water toward 7.5–8.5 even after you correct it. The fix is to **neutralise the bicarbonates first**, then fine-tune. You will see fizzing when you add acid — that is CO₂ releasing as the bicarbonates neutralise. Wait for the fizzing to stop before a final adjustment.

Always adjust pH **after** mixing nutrients, not before. And always **add acid to water — never water to acid**. Work in small amounts.

pH Down — Part A · Locally Foraged

The Dragoons and surrounding Cochise County provide several natural acid sources. These are best for minor daily adjustments on small systems. They are free, renewable, and satisfying to produce from your own landscape.

A1 — Prickly-Pear Vinegar

Prickly pear cactus (*Opuntia*) is abundant throughout the Dragoons and Sulphur Springs Valley. The fruit ferments readily into a mild fruit vinegar with a final pH of 3.0–3.5 — gentle enough for easy dosing on small systems.

What you need: ripe prickly-pear fruits (tuna) · a glass jar with a loose lid · a fine cloth strainer · a pH meter.

How to make it:

1. **Harvest.** Collect 10–15 ripe red/purple fruits using tongs. Rinse spines off under water.
2. **Mash.** Crush thoroughly in a glass jar, including skins and seeds — they carry the wild yeasts.

3. **Ferment.** Cover loosely with cloth (not airtight). Leave at room temp 2–3 weeks, stirring daily.
4. **Test.** After 2 weeks, strain a small sample and test pH. Target is 3.0–3.5.
5. **Strain.** Pour through fine cloth into a clean glass jar. Discard solids. Label and date.
6. **Store.** Refrigerate. Lasts 3–6 months; use within 1 week once opened at room temperature.

System size	Starting dose	Tip
1 gallon	1–2 ml, stir, wait 10 min, retest	Keep a small spray bottle for easy micro-dosing
3 gallons	3–5 ml, stir, wait 10 min, retest	Add slowly — prickly-pear vinegar is gentle but cumulative
5 gallons	5–8 ml, stir, wait 10 min, retest	Multiple small doses beat one large one

Prickly-pear vinegar adds very small amounts of natural fruit acids and trace minerals to your reservoir. At the low doses used for pH adjustment, this has negligible effect on the nutrient profile.

A2 — Fermented Citric Acid (from local fruit or honey)

Citric acid can be produced by fermenting sugar with *Aspergillus niger* mould, which grows naturally on overripe fruit skins in this region. This is, in fact, how industrial citric acid is manufactured — you are doing it at a small scale. The result is stronger than vinegar and more precise to dose.

1. **Prepare substrate.** Dissolve 2 tbsp local honey or prickly-pear juice in 1 cup warm water in a wide-mouth glass jar.
2. **Inoculate.** Add a small piece of overripe citrus peel or lemon skin — the white fuzz is *Aspergillus niger*.
3. **Ferment.** Cover with breathable cloth. Leave at 75–85 °F for 7–10 days. A white/grey mould mat will form on top.
4. **Check.** Test pH of the liquid beneath the mat — target is 2.0–3.0.
5. **Harvest.** Remove the mat and strain liquid through cloth. The clear acidic liquid is your pH down.
6. **Dilute.** Dilute 1:4 with water before use (1 part acid to 4 parts water) for easier dosing.

Do not consume the fermented citric-acid liquid — it is for plant use only. The *Aspergillus* mat is harmless in this context but should be discarded, not eaten. Work outdoors or with ventilation.

A3 — Sulfur-Dioxide Acid Water (advanced)

The Sulphur Springs Valley east of the Dragoons contains native sulphur deposits. Burning sulphur produces sulphur-dioxide gas, which dissolves in water to form sulphurous acid — a strong, effective pH down. This method is powerful and best reserved for pre-treating large volumes of hard well water.

1. **Collect sulphur.** Gather native sulphur from mineral deposits — yellow crystalline material from fumarolic areas.
2. **Set up outdoors.** Work completely outdoors, wind at your back. Never indoors — fumes are irritating.
3. **Burn a small amount.** Ignite a small piece in a metal dish. It burns with a blue flame and white fumes.
4. **Capture the gas.** Hold a bucket of water downwind of the burn so fumes dissolve into the surface.
5. **Test often.** Test the water pH every few minutes — it drops quickly. Stop at pH 4.0–5.0.
6. **Use immediately.** Sulphur-acid water is unstable — use within 24 hours. Do not store long-term.

Best use for this method: pre-treating a full reservoir of SE Arizona well water before adding nutrients. The strong acid neutralises bicarbonates efficiently. Then fine-tune with prickly-pear vinegar for the final pH adjustment.

pH Down — Part B · Homemade from Purchased Ingredients

If you want more consistency and precision than foraged options provide, these home-made options use inexpensive purchased ingredients to create reliable, stable pH-down solutions equivalent to commercial products. All ingredients are food-safe and widely available.

B1 — Citric-Acid Stock Solution ★ Best overall

Food-grade citric-acid powder is cheap (~\$5/lb), extremely stable, and makes a consistent pH down that closely mirrors commercial products. It does not add harmful ions to the reservoir and is the top recommendation for small hobby systems. A single pound will last a small grower 1–2 years.

1. **Purchase.** Buy food-grade citric-acid powder (not industrial). Grocery stores, Amazon, homebrew suppliers.
2. **Make stock solution.** Dissolve 1 tbsp (≈12 g) in 1 cup (240 ml) of clean water. Stir until fully clear.
3. **Label and store.** Pour into a dark glass bottle. Label: *pH Down — Citric Acid Stock*. Room temp, away from light.

4. **Dose.** Add 1 ml at a time. Stir, wait 5–10 min, retest before adding more.

System size	Starting dose	For bicarbonate pre-treatment
1 gallon	0.5–1 ml stock, retest	3–5 ml stock until fizzing stops, then fine-tune
3 gallons	1–2 ml stock, retest	8–12 ml stock until fizzing stops, then fine-tune
5 gallons	2–3 ml stock, retest	12–20 ml stock until fizzing stops, then fine-tune

Citric acid breaks down naturally in the reservoir over 3–5 days via microbial activity, so it does not accumulate. It leaves no residual ions that would affect the nutrient balance. Safe, clean, and effective.

B2 — White Vinegar Stock (budget option)

Standard white vinegar (5 % acetic acid) is available everywhere and works as a mild, easy-to-dose pH down. The main downside is that acetate ions can accumulate in recirculating systems over time. For small systems with regular reservoir changes every 7–14 days, this is not a significant issue.

1. **Purchase** standard white distilled vinegar, 5 % acidity. Any grocery brand.
2. **Use undiluted.** At 5 % it is already dilute enough for easy dosing.
3. **Dose directly.** Add 2–5 ml at a time. Stir, wait 10 min, retest.
4. **Reservoir changes.** Change every 7–10 days for fruiting crops to prevent acetate build-up.

System size	Starting dose	Notes
1 gallon	2–3 ml, stir, retest	Gentle — easier to control than stronger acids
3 gallons	5–8 ml, stir, retest	Multiple small additions beat one large dose
5 gallons	8–12 ml, stir, retest	Well-water pre-treatment may need 20–40 ml total

Avoid **apple-cider** vinegar — it contains organic compounds and sugars that promote bacterial growth in the reservoir. White distilled vinegar only.

B3 — Diluted Phosphoric Acid (closest to commercial pH Down)

Commercial pH Down products are typically 85 % phosphoric acid diluted to around 10–20 % concentration. You can buy food-grade phosphoric acid and make the same product at a fraction of the cost. Phosphoric acid also adds a small amount of phosphorus to the reservoir — a benefit for fruiting crops.

1. **Purchase.** Food-grade phosphoric acid 85 % (used in cola production). Homebrew and hydroponic suppliers online.
2. **Safety first.** Gloves and eye protection. The 85 % solution is a strong acid.
3. **Make a working solution.** Add 1 part phosphoric acid to 9 parts water, *slowly* (never water to acid). ~8.5 % solution.
4. **Label clearly.** *pH Down — Phosphoric 8.5 % — HANDLE WITH CARE*. Store away from children.
5. **Dose.** 0.5 ml at a time. Stronger than vinegar or citric acid — very small doses.

System size	Starting dose	Notes
1 gallon	0.2–0.5 ml, stir, retest	Very small doses — much stronger than vinegar
3 gallons	0.5–1.0 ml, stir, retest	Drop by drop with a syringe for best control
5 gallons	1.0–1.5 ml, stir, retest	Ideal for bicarbonate pre-treatment of well water

Option	Strength	Cost	Shelf life	Best used for	Rating
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pH Down — Full Comparison

Option	Strength	Cost	Shelf life	Best used for	Rating
A1 — Prickly-pear vinegar	Mild pH 3-3.5	Free	3-6 months refrigerated	Daily micro-adjustments. Minor tweaks on rainwater.	★★★★☆
A2 — Fermented citric acid	Medium pH 2-3	Free	Use within 1-2 weeks	Moderate corrections. Better for well water than vinegar.	★★★★☆
A3 — Sulphur acid water	Strong pH 1.5-3	Free	24 hrs only	Bicarbonate pre-treatment of large well-water volumes.	★★★★☆
B1 — Citric-acid stock ★	Medium pH 2-2.5	~\$5/lb (lasts 1-2 yrs)	12+ months sealed	Everything — best all-round option for hobby growers.	★★★★★
B2 — White vinegar	Mild pH 2.5-3	~\$2/bottle	2+ years sealed	Quick adjustments. Fine for rainwater systems.	★★★★☆
B3 — Phosphoric acid (diluted)	Strong pH 1-2	~\$15/bottle (lasts years)	2+ years sealed	Well-water pre-treatment. Closest to commercial pH Down.	★★★★★

pH Up — Part C · Locally Foraged

Most Dragoon Mountain growers adjust *down* far more often than *up* — bicarbonate-rich well water and nitrifying-bacteria activity both push pH downward over time. But rainwater-fed systems and long vegetative runs can drift acidic, and you will eventually

want a dependable pH-up option.

C1 — Oak-Ash Lye ★ Best local option

Oak woodland is the Dragoons' signature plant community, and oak ash — steeped in soft water — produces a potassium-rich alkaline liquid used historically for soap-making, tanning, and pickling. It is the most practical local pH-up source, and it has a useful side benefit: it adds potassium to your reservoir at the same time it raises pH.

What you need: fully-burnt oak ash from seasoned hardwood (no charcoal chunks, no resinous wood) · a non-metallic container · rainwater or RO water · fine cloth · a pH meter · gloves.

How to make it:

1. **Pack** a non-metallic container loosely with cold oak ash — a food-grade bucket or large glass jar works well.
2. **Percolate.** Pour 2–3 volumes of rainwater (or RO water) slowly through the ash, letting it drip through over several hours into a collection vessel. Do not use well water — its bicarbonates will interfere.
3. **Rest.** Let the lye settle overnight. It will be a dark amber-to-brown liquid; expect pH 11–13.
4. **Strain.** Pour through fine cloth into a labelled dark glass bottle. Discard the ash (it still has residual potassium — spread on compost, not in hydro).
5. **Store.** Seal tightly. Keeps indefinitely. Label: *pH Up — Oak Ash Lye — CAUSTIC*.

Oak-ash lye is **caustic**. Wear gloves and eye protection. Never splash on skin. Dilute heavily before reservoir use — this is the strongest pH-up option in the guide.

System size	Starting dose	Notes
1 gallon	0.1–0.3 ml lye, stir, wait 10 min, retest	Dilute lye 1:10 with water first for easier control
3 gallons	0.3–0.8 ml lye, stir, wait 10 min, retest	Drop by drop with a pipette
5 gallons	0.5–1.5 ml lye, stir, wait 10 min, retest	Pre-dilute to a 1:10 working solution and dose from that

Oak-ash lye adds potassium (K) — a macronutrient you want. This makes it the least disruptive pH-up choice for a nutrient programme: every correction doubles as a small top-up of K. The one crop where this matters most is fruiting peppers and tomatoes, which benefit from extra potassium in the flowering and sizing windows anyway.

C2 — Hydrated Lime from Caliche or Limestone

Caliche (calcium carbonate hardpan) and limestone outcrops throughout Cochise County can be processed into hydrated lime — calcium hydroxide — which is a strong pH-up material that also contributes calcium. It is more work than oak-ash lye and less kind to the skin, but it is a useful backup.

1. **Burn limestone or caliche** in a hot fire (bonfire or kiln) for several hours. This drives off CO₂ and leaves quicklime (CaO).
2. **Slake the quicklime.** Slowly add a small amount of water to the hot material — it will hiss and steam. Keep adding water until it forms a smooth paste of calcium hydroxide (hydrated lime).
3. **Dilute.** Stir a teaspoon of paste into 1 cup of water. Let settle. The clear liquid above the sediment is “lime water” — pH 12 +.
4. **Dose** the clear supernatant drop by drop — this is strong.

Quicklime reacts violently with water and can burn skin severely. Protective gear is non-optional. If you can avoid this method with oak-ash lye, do.

System size	Starting dose (lime water supernatant)	Notes
1 gallon	0.5–1 ml, stir, retest	Very strong — tiny amounts only
3 gallons	1–2 ml, stir, retest	Pre-dilute; drop by drop
5 gallons	2–3 ml, stir, retest	Useful if you are calcium-deficient as well as acid

pH Up — Part D · Homemade from Purchased Ingredients

D1 — Potassium Hydroxide Stock (closest to commercial pH Up)

Commercial pH Up is typically a dilute potassium-hydroxide solution. Food-grade potassium hydroxide (KOH) pellets — the same material used in soap-making — are widely available from soap-supply shops and some homebrew suppliers. As with phosphoric acid, you can make the exact commercial product at a fraction of the cost.

1. **Purchase** food-grade KOH pellets (often labelled “potassium hydroxide food grade” or “caustic potash”). Store dry, sealed.
2. **Safety first.** Gloves, eye protection, ventilation. KOH pellets are strongly caustic and hygroscopic.
3. **Make stock.** Slowly add 1 tbsp (≈10 g) of KOH pellets to 1 cup (240 ml) of distilled water — the solution will warm noticeably. Stir with a glass or plastic utensil (not metal) until fully dissolved.

4. **Bottle and label.** Dark glass or thick HDPE bottle. Label: *pH Up — KOH 4 % — CAUSTIC*.
5. **Dose.** 0.3–0.5 ml at a time. Stir thoroughly, wait 10 minutes, retest.

System size	Starting dose	Notes
1 gallon	0.1–0.3 ml, stir, retest	Like oak-ash lye, this also adds useful potassium
3 gallons	0.3–0.8 ml, stir, retest	Drop by drop
5 gallons	0.5–1.5 ml, stir, retest	Clean, consistent, commercial-equivalent

D2 — Baking-Soda Solution (budget, temporary)

Sodium bicarbonate (baking soda) raises pH gently and is safe to handle, but it adds sodium — which competes with potassium and calcium uptake and accumulates in recirculating systems. It is fine as an occasional emergency fix, but not as a regular pH-up strategy.

System size	Starting dose (½ tsp baking soda in 1 cup water, then dose the liquid)	Notes
1 gallon	2–5 ml of solution, stir, retest	Change reservoir within the week to flush Na
3 gallons	5–15 ml, stir, retest	Not for long runs
5 gallons	10–25 ml, stir, retest	Consider oak-ash lye or KOH stock instead

pH Up — Full Comparison

Option	Strength	Cost	Shelf life	Best used for	Rating
C1 — Oak-ash lye ★	Strong pH 11–13	Free	Indefinite sealed	The Dragoon default pH-up. Adds useful potassium.	★★★★★
C2 — Hydrated lime (caliche)	Strong pH 12+	Free	Indefinite dry	Calcium-plus- pH-up when deficient in both.	★★★☆☆

Option	Strength	Cost	Shelf life	Best used for	Rating
D1 — KOH stock	Strong pH 13+	~\$10/lb (lasts years)	12+ months sealed	Commercial-equivalent precision. Adds K.	★★★★★
D2 — Baking soda	Mild pH 8-9	~\$1/box	2+ years sealed	Occasional emergency only — adds sodium.	★★☆☆☆

Universal Dosing Protocol — Any pH Adjuster

1. **Fill reservoir.** Start with your water source. Note starting pH and EC with your meters.
2. **Bicarbonate check.** If using well water, add acid slowly until fizzing stops completely. This neutralises bicarbonates.
3. **Add nutrients.** Masterblend, then Epsom, then calcium nitrate — into treated water.
4. **Check pH.** Measure after nutrients are dissolved. It will likely be 6.5-7.5.
5. **Dose.** Add chosen pH adjuster 1 ml at a time. Stir thoroughly after each addition.
6. **Wait.** 5-10 minutes between additions — pH continues to shift after mixing.
7. **Target.** Stop when pH reads 5.8-6.2. Your sweet spot for most crops.
8. **Daily check.** Every morning. Small doses as needed.

Recommended setup for Dragoon Mountain hobby growers:

- **Citric-acid stock (B1)** — precise daily pH-down adjustments and fine-tuning after nutrients are added.
- **Prickly-pear vinegar (A1)** — minor corrections, made fresh from the landscape each season.
- **Oak-ash lye (C1)** — your pH-up, made fresh each year from last winter's fireplace.

If your well water is very hard (pH 8+), add **phosphoric acid (B3)** for the initial bicarbonate pre-treatment before switching to citric acid for fine-tuning. This two-stage approach gives the most stable, predictable pH management with minimal effort.

Closing

Every acid and alkali in this guide has a commercial equivalent sitting on a hydroponic-supply shelf somewhere. The point of making your own is partly economic — citric acid powder at \$5 per pound will outlast most systems — and partly about tying the reservoir back into the landscape it sits in. Oak ash from a winter fire makes the pH-up that corrects the drift on a fruiting pepper in August. Prickly-pear fruit from a hillside in September

ferments into the vinegar that trims a rainwater reservoir in November. Caliche dissolved in that vinegar is the calcium that stops blossom-end rot in the tomatoes.

Measure everything. Keep both meters calibrated. And keep two bottles of each kind on hand — one on the bench, one as the backup — because the one moment you absolutely need a working pH-down dropper is the moment the bottle is empty.